

SC2000/CZ2100 Probability & Statistics

Part 1 Instructor: Dr Michele Nguyen

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Part 2 Instructor: A/P Adams Kong

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~ 24 Sessions of lectures

~ 12 Sessions of tutorials

50 min per session

Assessments:

- 4 Quizzes – 25% each
- No Final exam

Part 1 quizzes

Quiz 1: Feb 16

Quiz 2: Mar 1

Part 1 (Week 1-7):

Ch1 Introduction to Statistics

Ch2 Presenting Data

Ch3 Summarizing Distributions

Ch4 Bivariate Data

Ch5 Probability Theory

Ch6 Probability Distribution

- Random Variables
- Discrete Distribution
- Continuous Distribution

Part 2 (Week 8-13):

Statistical Inference to be taught by Prof Adams
Kong

- TEL supporting Materials for Ch1-4 (short video clips and notes) are available in NTULearn.

Students are advised to watch the video clips before attending the lectures for Part 1 (Ch1-4).

- Lectures (Week 1 to 7)
 - Discussion & worked examples on topics covered in the TEL materials
 - Additional topics not covered in TEL materials (particularly those in Ch 5 and 6)
 - Recorded lectures will be posted in NTULearn Course Media

- Recess week (Self-study non-exam)
 - Use of R programming for analysis and presentation of statistical data (Practice materials will be available in NTULearn)
- Tutorial – One session per week, to begin in Week 2

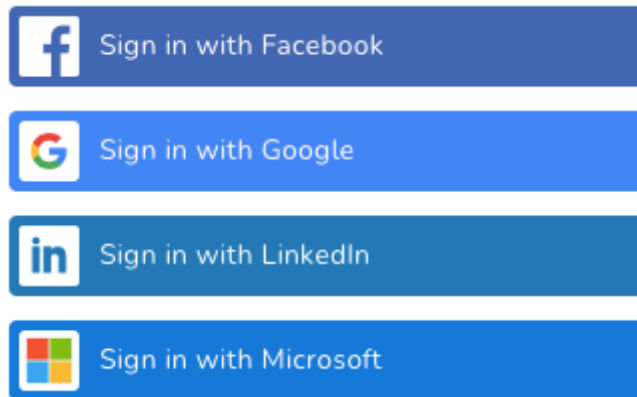
All Course Materials are available in [NTULearn](#).

Prof Kong will take over the teaching from Wk 8.

Students participation during lectures using Student Response System - Wooclap

A QR code or URL will be presented for students to join
Wooclap.

Scan the QR code with your mobile phone and access
the web page. If need be, you can sign in with your
NTU email or any of the following:



Let's try out the Wooclap session by answering this question using your mobile phone:

Have you taken a course on Probability and Statistics before, eg: in JC or Poly?



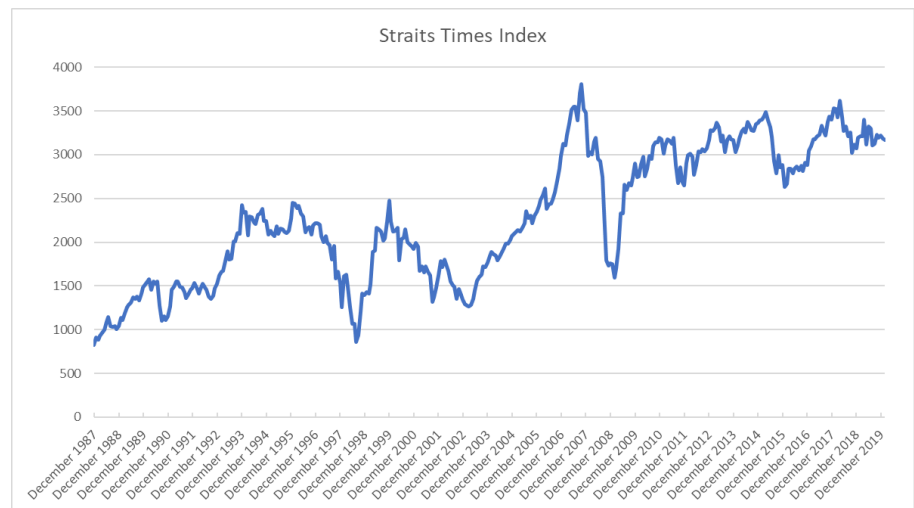
Ch 1. Introduction to Statistics

- Descriptive & Inferential Statistics
- Types of Variables
- Types of Measurement Scale
- Linear Transformations
- Distributions
- Percentiles

Statistics involve gathering, organizing, analysing, interpreting and presenting data.

Statistics are being used everywhere:

- Number of students enrolled in this course
- Index measuring stock market



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- Index measuring stock market
- Singapore household income
- Live data of new Covid-19 cases

Statistics involve gathering, organizing, analysing, interpreting and presenting data.

Statistics are being used everywhere:

- Number of students enrolled in this course
- Index measuring stock market
- Singapore household income
- Live data of new Covid-19 cases
- Opinion poll, benchmark poll, tracking polls, etc.

Statistics provide information to make decisions.



E.g. Insurance companies use past car accident statistics to decide the car insurance premiums.

Statistics are used to make claims.

Example: A toothpaste manufacturer claims that more than 80% of Dentists recommend a particular brand of toothpaste. This was based on surveys of dentists which allow selection of one or more brands.

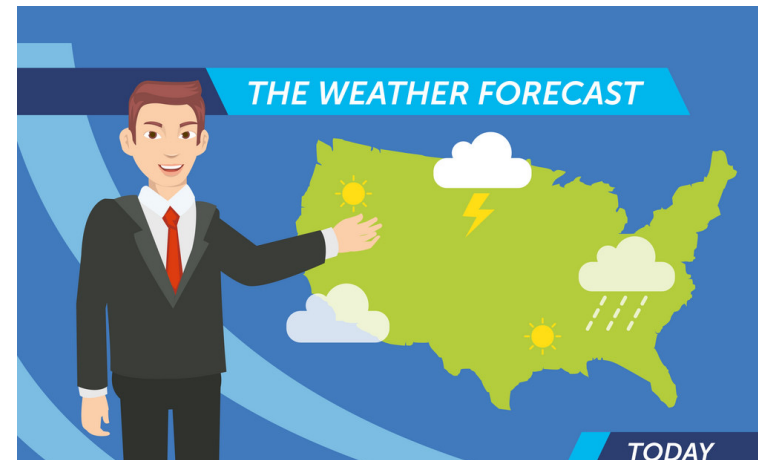
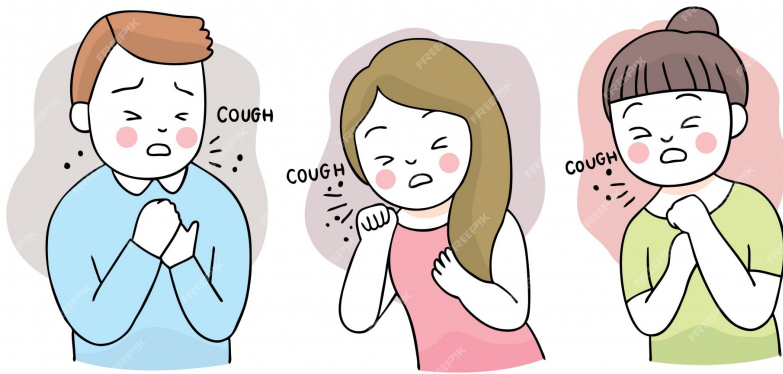
Is the claim misleading?

Yes.

Because it may be understood that 80% of dentists recommend this brand over the others. It should be noted that other brands were also recommended and may be as much as that particular brand.

Why is Statistics important?

- Predicting the spreading of diseases
- Weather forecasting based on statistics



Why is Statistics important?

- Predicting the spreading of diseases
- Weather forecasting based on statistics
- Provide informed choice on investment decision
- AI or machine learning based on past statistical data



Frequently bought together



These items are shipped from and sold by different sellers. [Show details](#)

- ☒ This Item: Nikon D810 FX-format Digital SLR Camera Body \$2,796.95
- ☒ Nikon MB-D12 Multi Battery Power Pack \$280.66
- ☒ Lexar Professional 1000x 128GB SDXC UHS-II/U3 Card (Up to 150MB/s read) w/Image Rescue 5 Software... \$96.83

Customers who bought this item also bought



■ Descriptive & Inferential Statistics

Descriptive statistics – summarize and describe important features of the data collected. Does not generalize beyond the data collected.

Inferential statistics - collect samples to draw inferences about the population, i.e. formal guesses of statistical parameters about the population by looking at the samples.

A teacher wishes to know whether the males in his class have more conservative attitudes than the females. A questionnaire is distributed assessing attitudes and the males and the females are compared.

Is this an example of descriptive or inferential statistics?

The screenshot shows a Wooclap poll interface. At the top, a red banner reads "You cannot vote anymore". The main text of the poll is: "A teacher wishes to know whether the males in his class have more conservative attitudes than the females. A questionnaire is distributed assessing attitudes and the males and the females are compared." Below this, the question is: "Is this an example of descriptive or inferential statistics?". There are two answer options: "Descriptive statistics" (highlighted in green) and "Inferential statistics" (highlighted in blue). A white box with a play button icon is overlaid on the question text, and a white arrow points to it. The interface also shows a progress bar at the bottom with the Wooclap logo, a search icon, "100 %", "33% correct", and "9 / 13" questions.

A cognitive psychologist is interested in comparing two ways of presenting stimuli on subsequent memory. Twelve subjects are presented with each method and a memory test is given.

What would be the roles of descriptive and inferential statistics in the analysis of these data?

Descriptive statistics – we describe and analyze the data from the sample.

Inferential statistics – we use the data from the sample to generalize to a larger population of people.

■ Types of Variables

In statistics, we can broadly group variables into two categories:

Quantitative – Numeric and represent measurable quantities.

Qualitative – Non-numeric and represent categories/groups.

Are these qualitative or quantitative?

- | | |
|--|--------------|
| 1. Cities where you have visited | Qualitative |
| 2. Movie rating (1 star, 2 stars, etc.) | Quantitative |
| 3. Temperature | |
| 4. The amount of cash you have in your pocket (50cts, \$1, etc.) | |

■ Types of Variables

For quantitative variables, we can further differentiate between:

Discrete variables

Continuous variables

■ Types of Variables

For quantitative variables, we can further differentiate between:

Discrete variables – discrete points on a scale

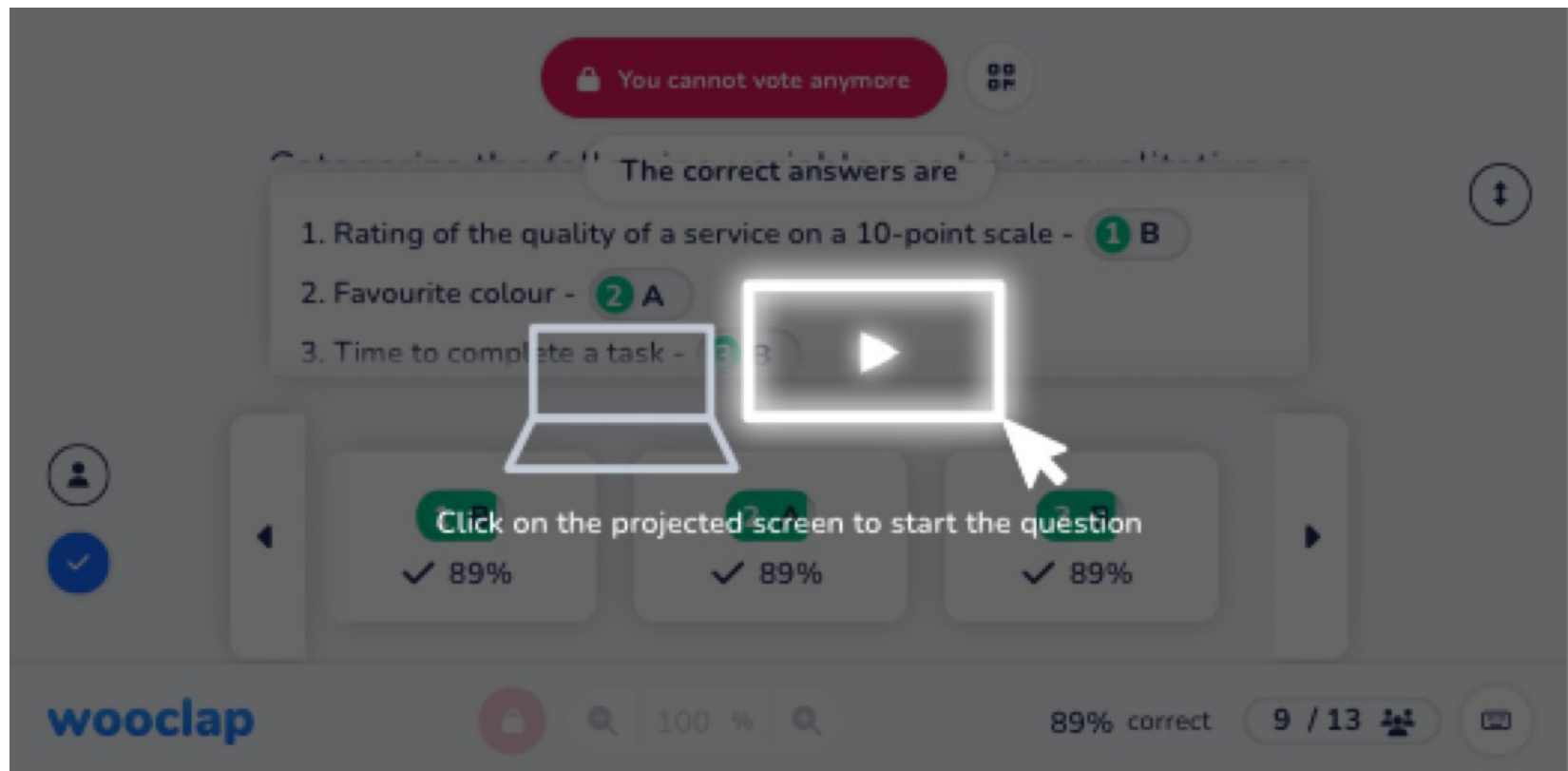
E.g. Number of times we visited different cities

Continuous variables – continuous scale and not discrete steps

E.g. Amount of money spent in each city

■ Types of Variables

Categorize the following variables as being qualitative or quantitative.



■ Types of Measurement Scales

Four basic levels of measurement scales:

1. Nominal – names or labels with no specific order

E.g. Cities where you have visited

2. Ordinal – variables in a specific order

E.g. Movie rating

■ Types of Measurement Scales

3. Interval – numerical scales in which intervals have the same interpretation throughout, but no true zero

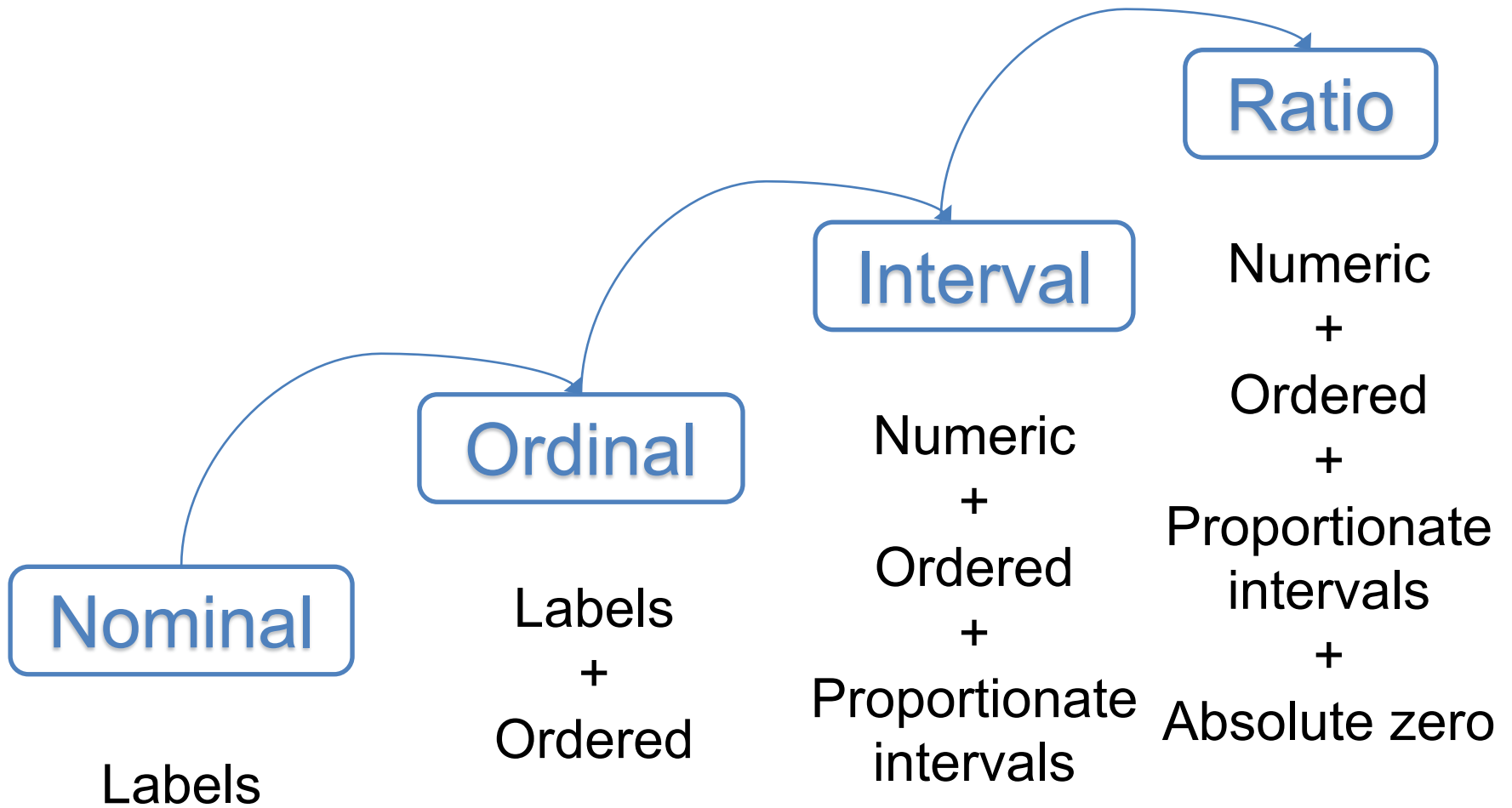
E.g. Temperature

4. Ratio – include all the characteristics of interval scale, plus it has zero position indicating the absence of the quantity being measured

E.g. The amount of cash you have in your pocket

Specify the measurement scales used for the following variables:





■ Linear Transformations

Transform data from one measurement scale to another.

Examples:

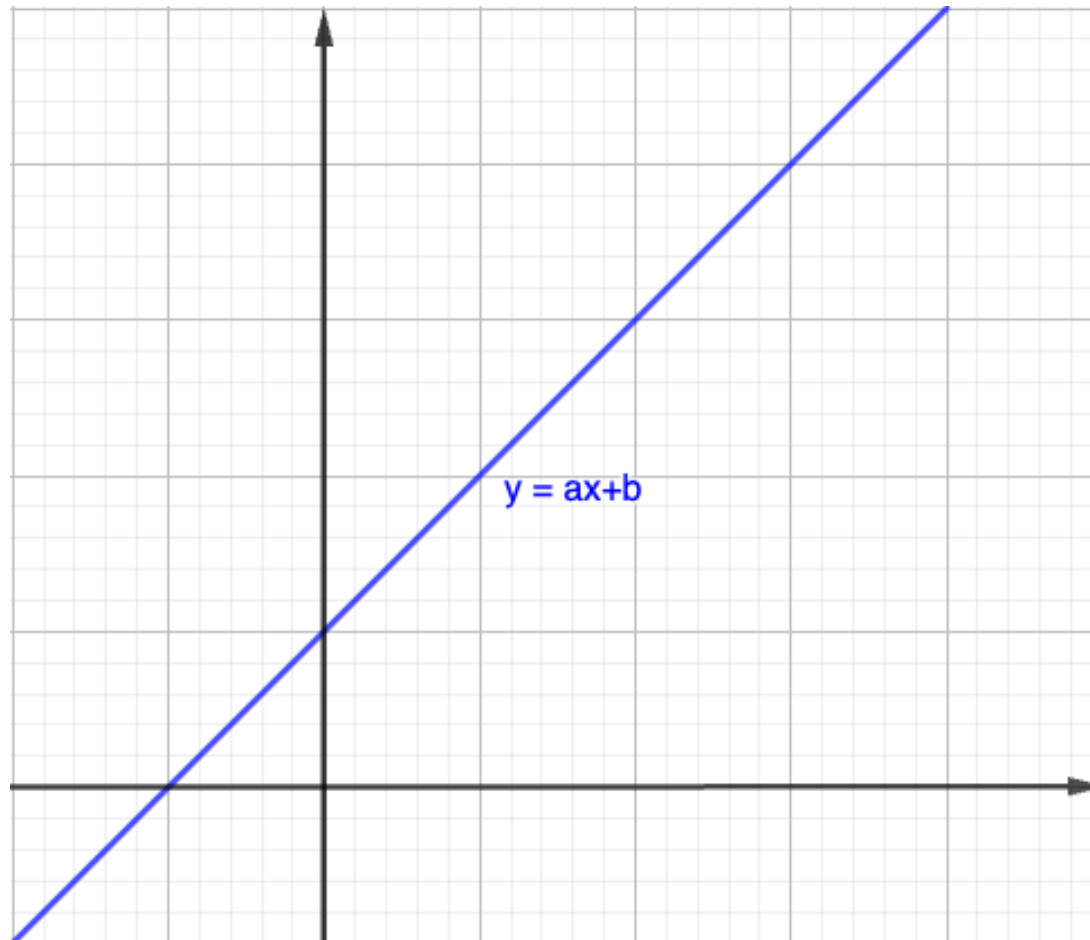
Convert length measured in X feet to measurement in Y metres, i.e.

$$Y = 0.3048 X$$

Convert temperature in Fahrenheit to Centigrade:

$$C = 0.5556 F - 17.778$$

■ Linear Transformations



■ Linear Transformations

Which of the following are linear transformations?

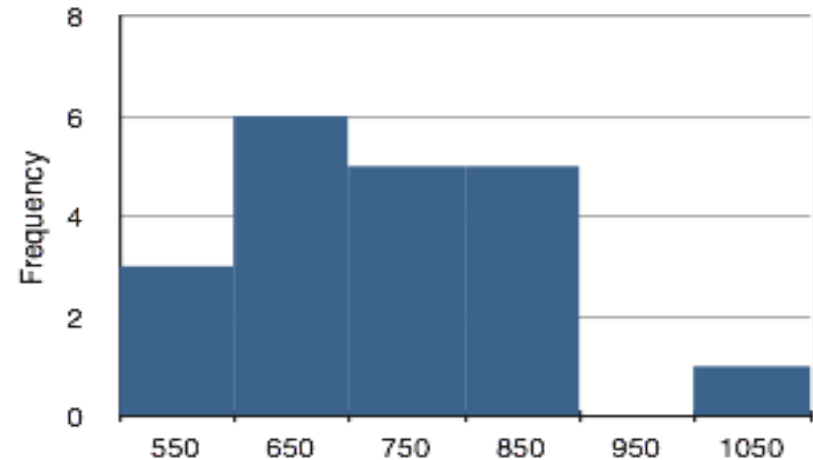
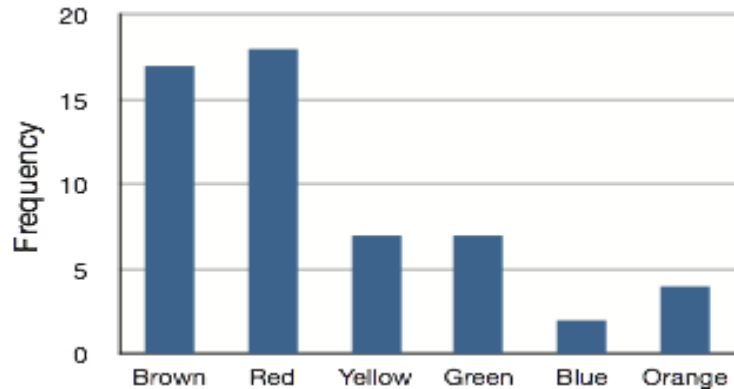


CX2100 Probability & Statistics

Ch 1. Introduction to Statistics

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- Types of Variables
- Types of Measurement Scale
- Linear Transformations
- **Distributions**
- **Percentiles**

■ Distributions



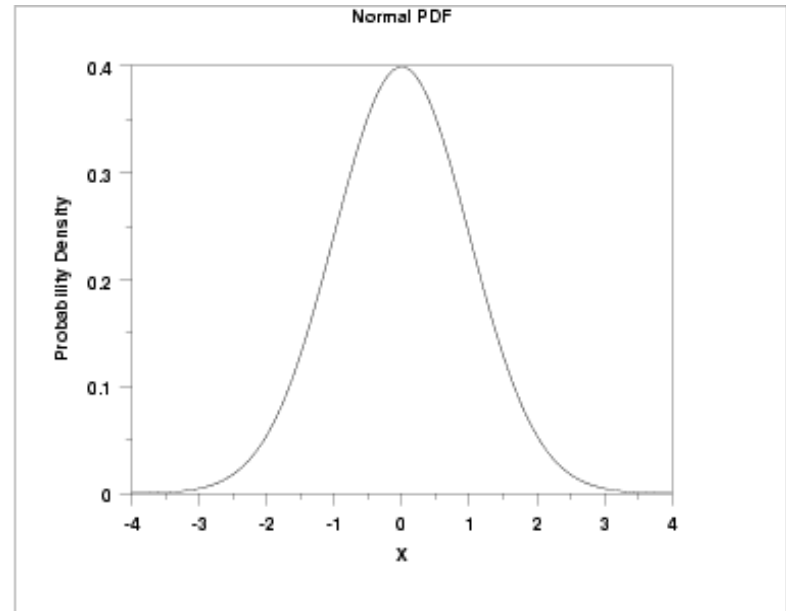
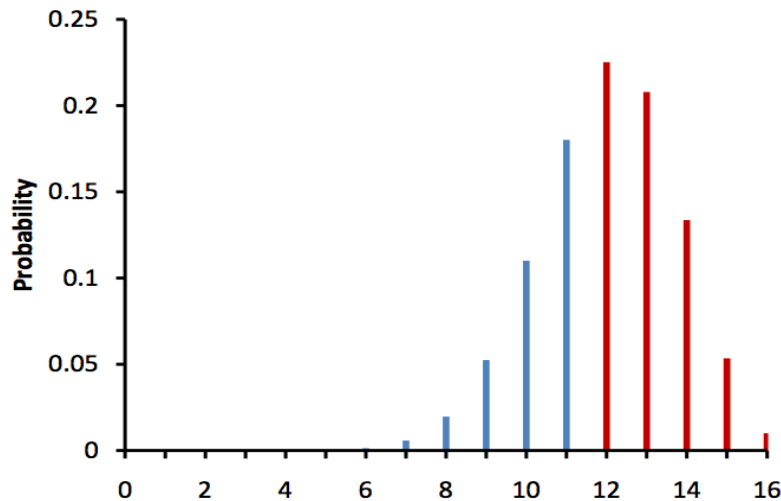
Frequency distribution

- Discrete/qualitative variables
- Continuous variables

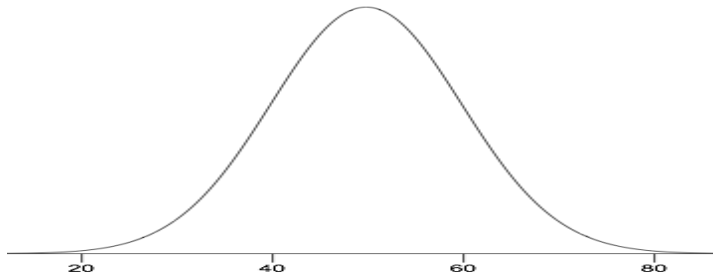
■ Distributions

Probability distribution

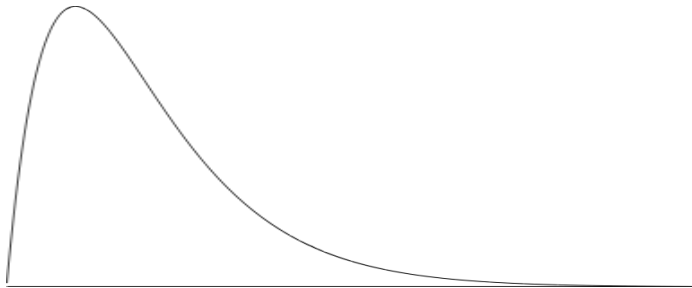
- Discrete variable:
Probability mass function (pmf)
- Continuous variable:
Probability density function (pdf)



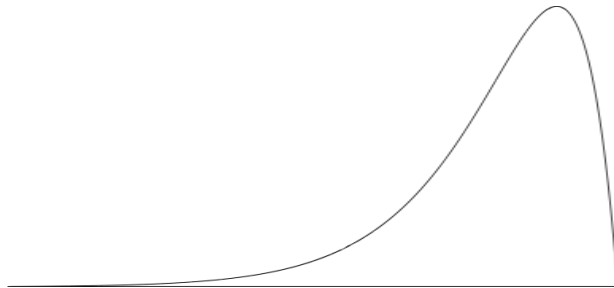
■ Distributions



Symmetric



Positive skew (to the right)

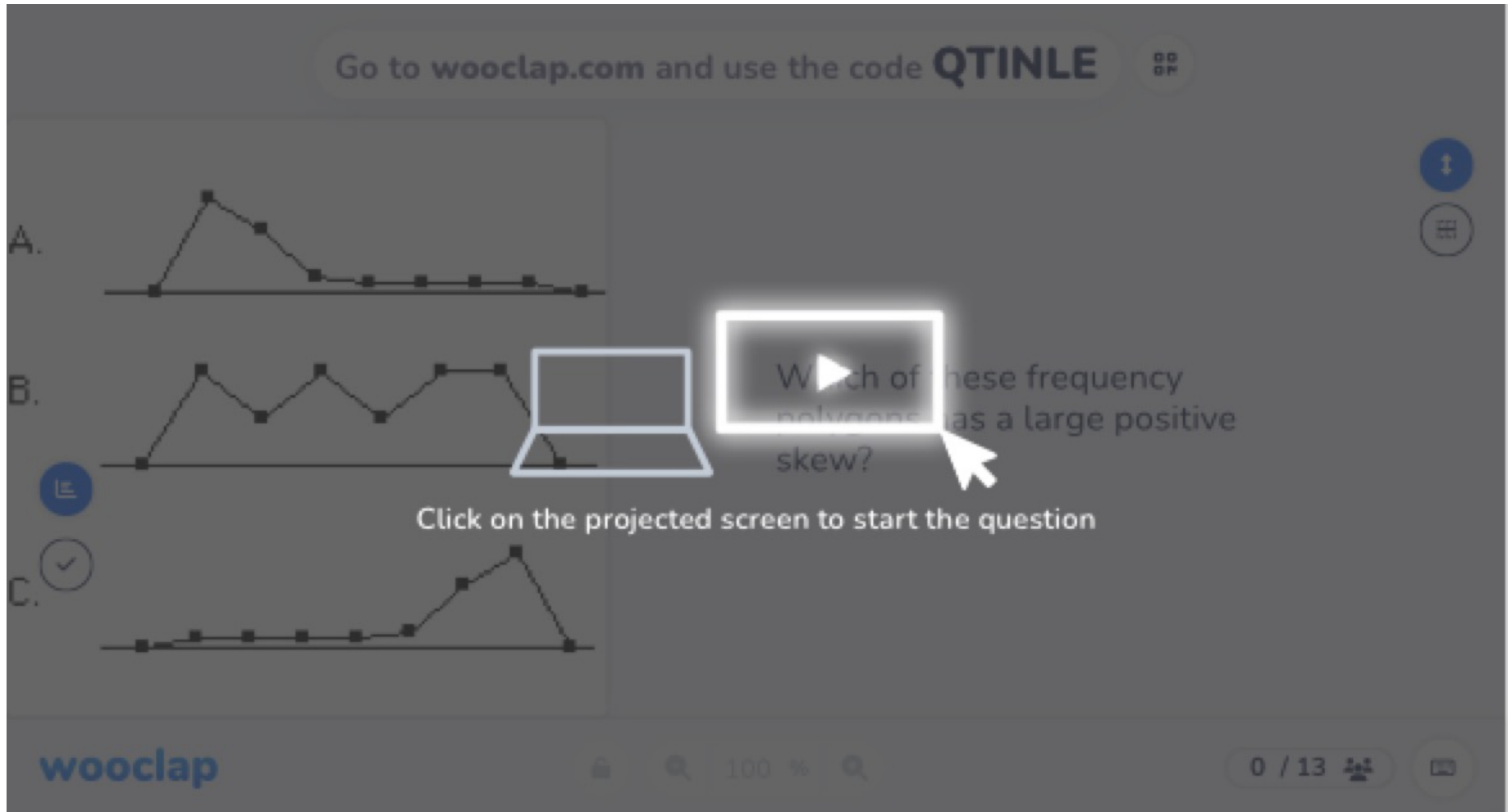


Negative skew (to the left)

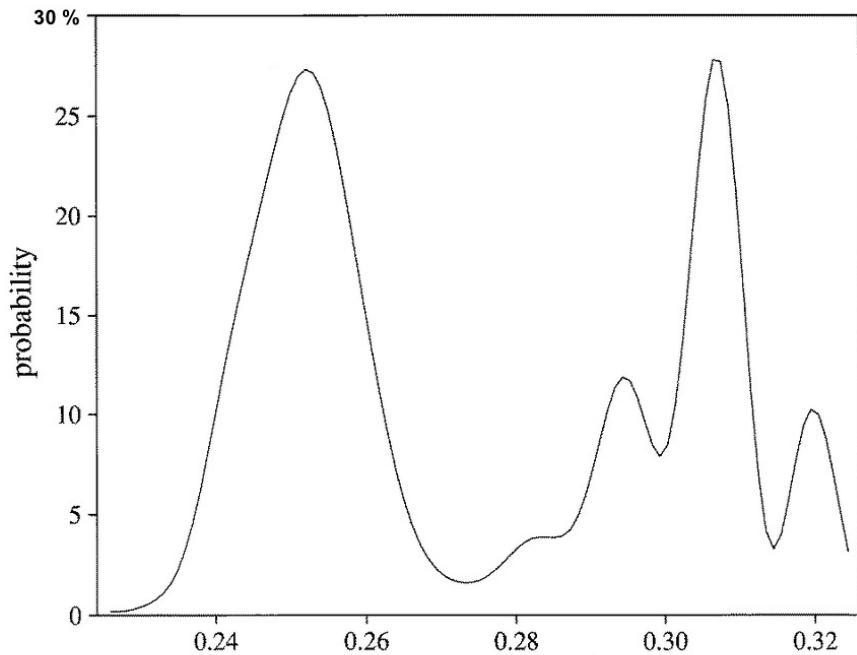
Shapes of distributions

- symmetric
- skewed to the right
- skewed to the left

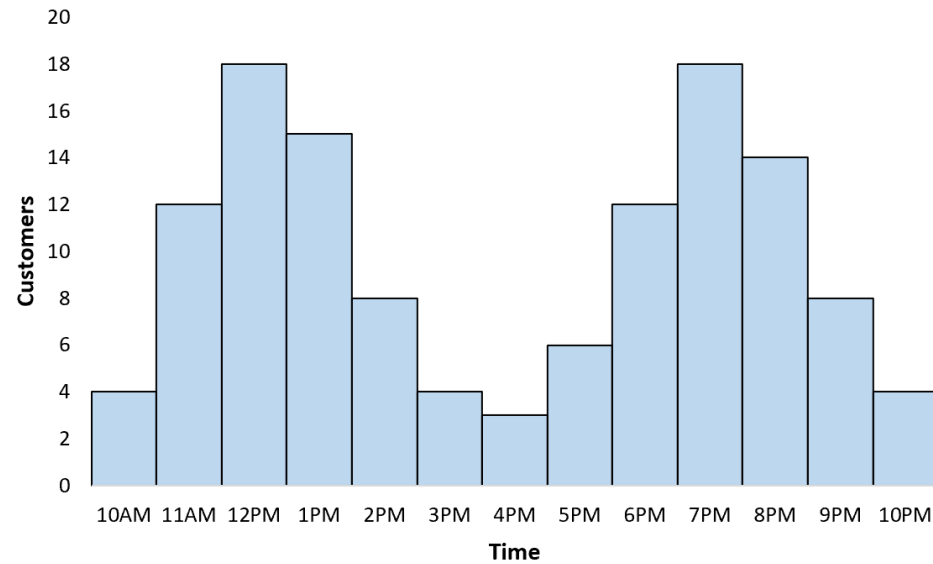
Which of the following frequency polygons has a large positive skew?



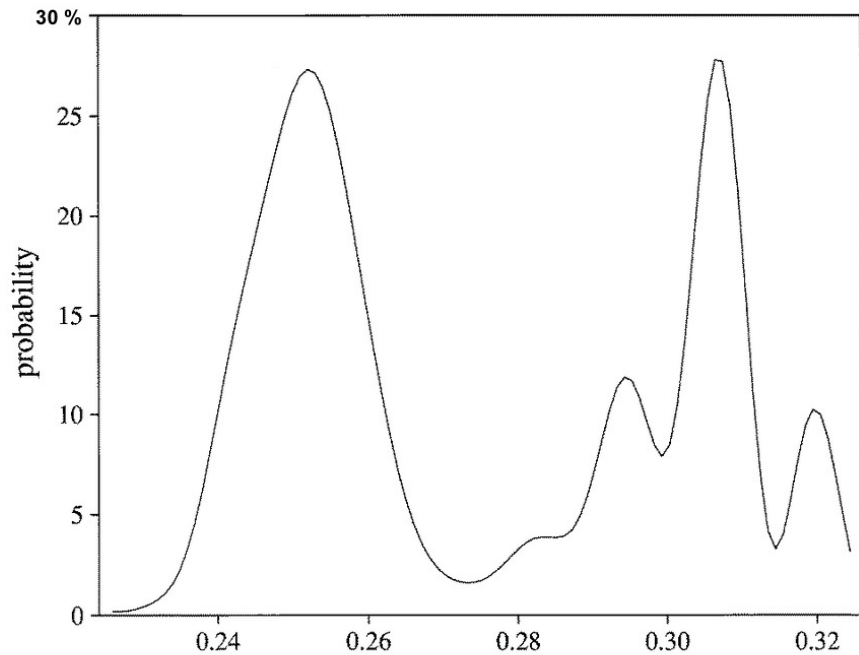
■ Distributions – other shapes:



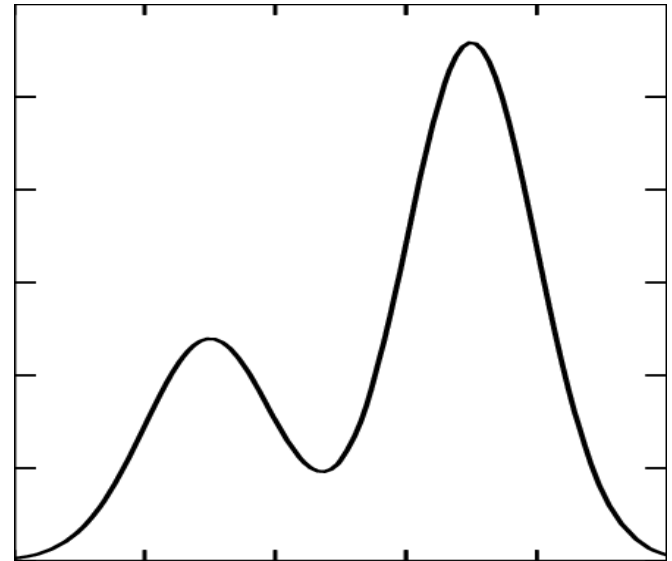
Multi-modal distribution



■ Distributions – other shapes:



Multi-modal distribution



Bimodal distribution

■ Percentile

In certain experiments, it is more meaningful to compare the outcomes obtained.

Eg: if you know that your quiz marks is 65 out of 100, you may not know how well you have done compared to others in your class.

A **percentile** is a comparison score between a particular score and the scores of the rest of a group.

■ Percentile - calculation

Calculating P^{th} Percentile for a set of N data:

1. Compute the rank $R = \frac{P}{100} \times (N + 1)$
2. Let $I_R = \text{Integer part of } R$ and $F_R = \text{Fractional part of } R$

■ Percentile - calculation

Calculating P^{th} Percentile for a set of N data:

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Data arranged in the order of magnitude

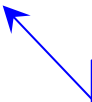


2. Let I_R = Integer part of R and F_R = Fractional part of R

3. P^{th} Percentile = Data at rank I_R +
(Data at rank $(I_R + 1)$ – Data at rank I_R) $\times F_R$

Given data: [3, 5, 7, 8, 9, 11, 13, 15], compute the 25th and 75th percentile.

1. Compute the rank $R = \frac{P}{100} \times (N + 1)$



Data arranged in the order of magnitude

2. Let I_R = Integer part of R and F_R = Fractional part of R

3. P^{th} Percentile = Data at rank I_R +
(Data at rank $(I_R + 1)$ – Data at rank I_R) $\times F_R$

Compute the 25th percentile for the data set of 20 values:

1	2	3	4	5	5	6	6	6	7
7	8	8	8	9	9	9	10	10	10



Click on the projected screen to start the question

Practical example:

Graduate Management Admissions Test - a standardized test for application to graduate-level business programs.

Report online

UNOFFICIAL GMAT® SCORE REPORT

	Scaled Score	
Integrated Reasoning	6	
Quantitative	47	
Verbal	47	
Total*	750	

total score is derived from your Verbal and Quantitative scores.

Official Score Report, including your Analytical Writing Assessment score, integrated score and percentile ranking.

Example: Percentile for benchmarking.

Suppose Jimmy earns a monthly wage of \$6,200. He would like to compare his wages with those working in the same occupation and industry.

